

HSE | 2025 | Econometrics

Women in National Legislatures and Health Expenditures

executed by the students of group 2301

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How Women's Representation in National Parliaments is Associated with Government Health Spending?

Hypothesis I

Greater *representation* of women in national parliaments is associated with a higher proportion of *health-related spending*.

Hypothesis II

The positive association between women's *representation* in national parliaments and *health-related spending* is stronger in countries with higher levels of *economic development*.

Female legislators prioritise social spending on health, education, and social welfare, redistributing government funds towards society.

- Historically, female policymakers have focused on social protection and labor integration (Creese, 1988)
- Women in politics show greater sensitivity to social cues and welfare issues (Croson, 2009).
- Empirical studies link women's political representation to higher public health spending (Mavisakalyan, 2012)
- The influence of female parliamentary representation on health expenditures is conditional on a country's level of economic development .

Data and Measurement: Key Variables

Data Source

World Bank (2000-2018)

Sample

161 countries, 2 885 obs.

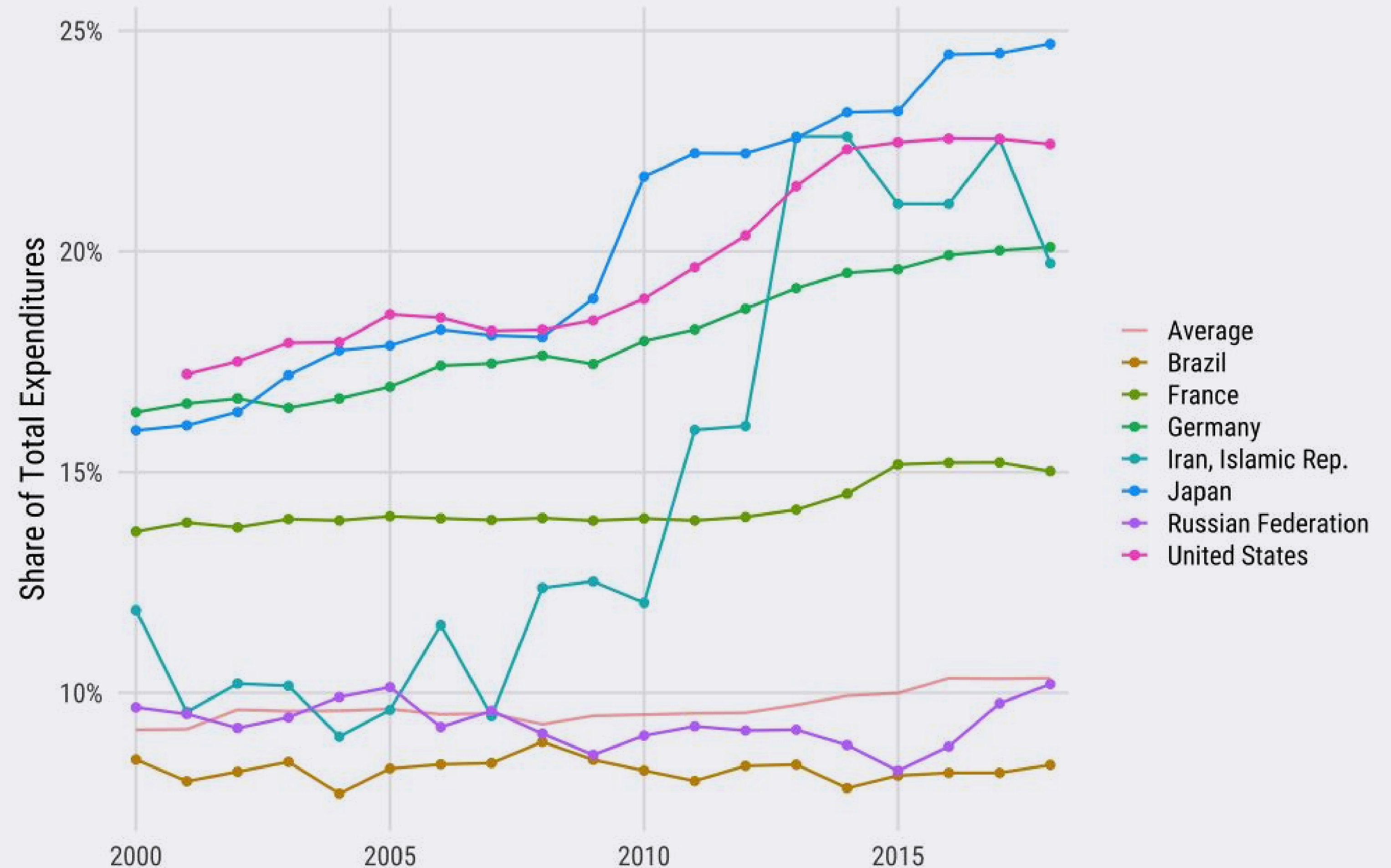
Dependent Variable

Domestic general government health expenditure (% of general government expenditure)

Independent Variables

Proportion of seats (%) held by women in national parliaments, GDP per capita (\$B)

State Public Health Expenditure over Time



Data and Measurement: Control Variables

Control variable	Theoretical Link to Women in Parliament	Theoretical Link to Health Expenditure
Political regime (Democracy Index)	Democratic systems <i>create pathways</i> for women's representation	Democratic governments are <i>more responsive</i> to public demands
Urbanization (% of urban population)	Cities promote more egalitarian views and give <i>more political opportunities to women</i>	In cities, the <i>quality</i> of services and accessibility of healthcare <i>are higher</i>
Female economic integration (% of female labour force)	Women's participation in the economy is a prerequisite for their political representation	Women's participation in the economy <i>expands the financing of health insurance</i>
Life expectancy at birth, total (years)	High life expectancy linked to <i>slower women's political progress</i> due to conservative values	High life expectancy leads to an aging population, <i>increasing the demand for medical services</i>

Results: Hypothesis I

Relation between the proportion of representation of women in parliaments and the share of health expenditures is not linear or universal.

	(1)	(2)	(3)	(4)	(5)
Women in Parliament (%)	0.135***	0.058**	0.027*	0.070***	0.007
Log GDP per capita		0.363	−0.044	0.561	−0.449
Urban population (%)		0.022	0.033	0.013	−0.030
Democracy index		0.018	−0.009	0.016	−0.008
Life expectancy (years)		0.190***	0.011	0.186***	−0.076
Female labor force (%)		0.079***	0.181***	0.074***	0.129*
Constant	7.229***	−12.172***			
Observations	2868	2868	2868	2868	2868
R2	0.102	0.390	0.882	0.404	0.887
Within R ²			0.045	0.401	0.027
Country FE	No	No	Yes	No	Yes
Year FE	No	No	No	Yes	Yes
SE type	Het.-rob.	by state	by state	by state	by state

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Conclusion of the linear analysis

Without accounting for one-way or two-way fixed effects, we get a positive association.

With two-way fixed effects, the linear relationship becomes statistically insignificant, indicating no universal impact.

This leads us to considering conditional relationship.

Results: Hypothesis II

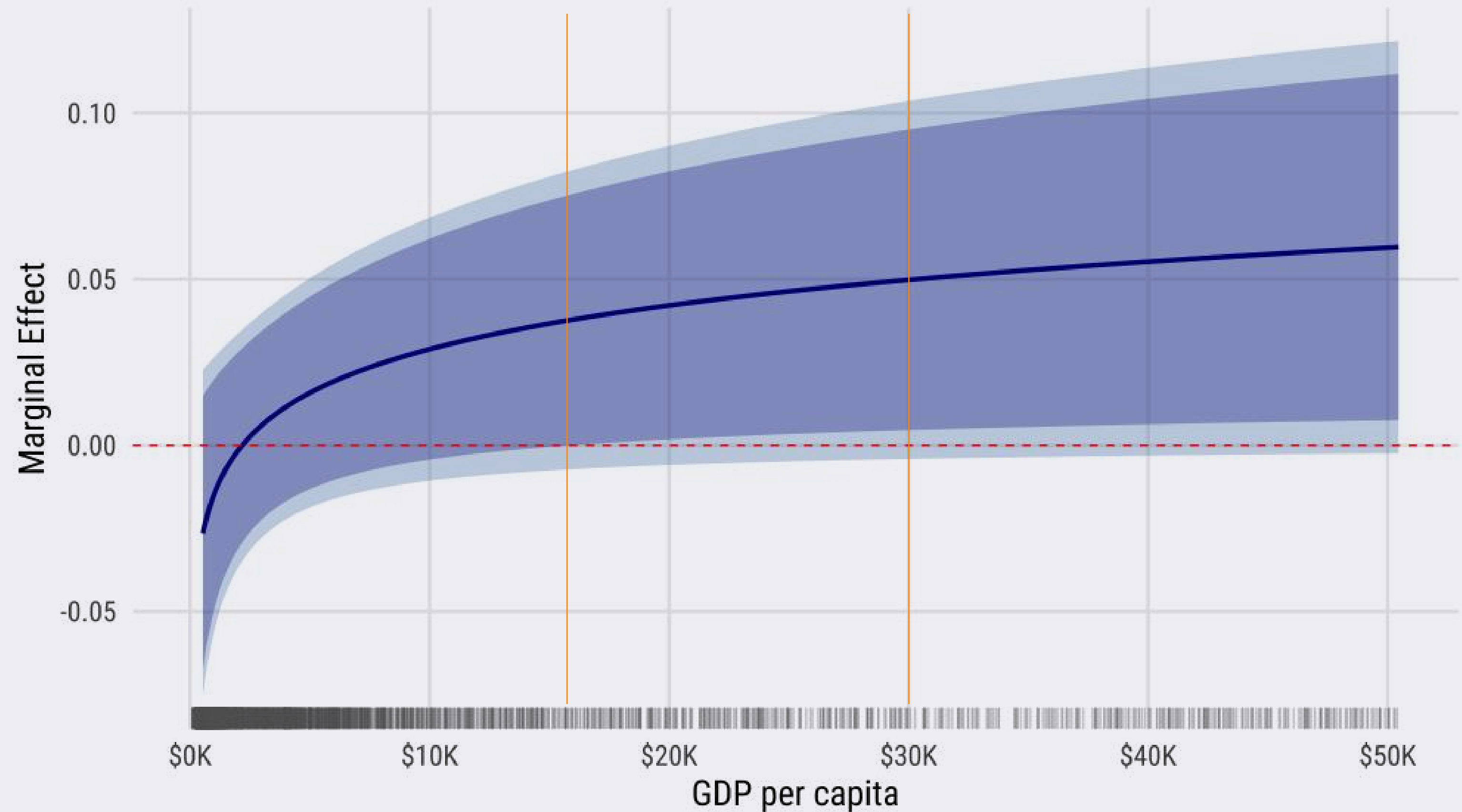
Women's influence is wealth-dependent: GDP per capita may condition the ability of female legislators to influence budgetary priorities.

Within low-income countries
no statistically significant effect

Within middle-income countries
1-point $\uparrow$ in women's parliamentary share $\rightarrow$ 0.03–0.05-point $\uparrow$ in health expenditure share

Within higher-income countries
1-point $\uparrow$ in women's parliamentary share $\rightarrow$ $\sim$ 0.06-point $\uparrow$ in health expenditure share

Blue: 95% CI | Dark blue: 90% CI



Limitations and Further Research

Future Research Directions

The use of instrumental variables (e.g., the introduction of gender quotas) or regression gaps to establish causation.

Expansion of measurements:

- The study of absolute expenses and spending efficiency.
- Taking into account the real political influence of women (the power index).
- The use of alternative indicators of fiscal space.

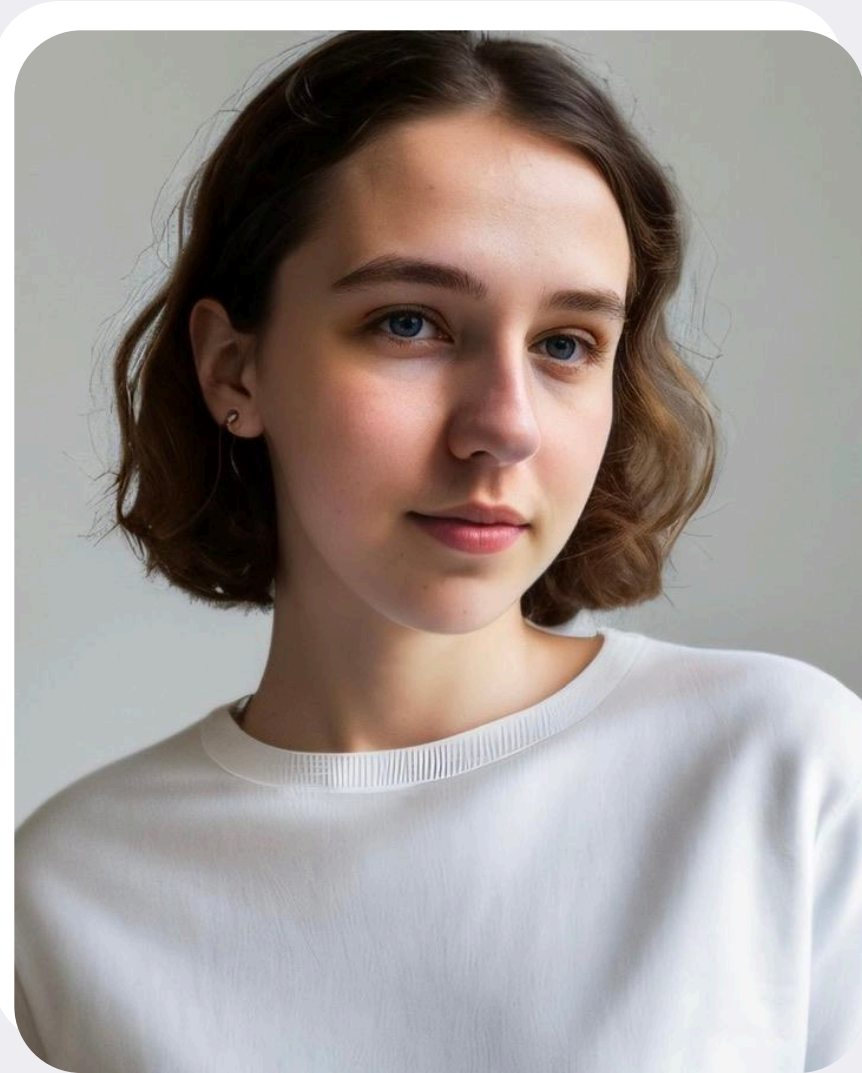
Capturing long-term and Institutional change, using longitudinal analysis over decades to trace the evolution of healthcare systems.

Key limitations of our research

- ➔ **Omitted Political and Institutional Factors**
Our data cannot account for key political mechanisms that likely influence both representation and spending.
- ➔ **Reverse Causality**
The analysis raises a question on the direction of the relation.
- ➔ **Western Bias**
Our core assumptions about how representation translates into policy are largely derived from OECD contexts.
- ➔ **Unobserved Informal Dynamics**
We measure formal seat shares, not the actual power or influence women wield within political systems.



Our Team



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Appendix: Effect of Women’s Parliamentary Representation on Health Expenditure

	(1)	(2)	(3)	(4)	(5)	(6)
Women in Parliament (%)	0.135*** (0.008)	0.058** (0.026)	0.027* (0.015)	0.070*** (0.026)	0.007 (0.017)	−0.146* (0.082)
Log GDP per capita		0.363 (0.394)	−0.044 (0.265)	0.561 (0.412)	−0.449 (0.319)	−0.646* (0.330)
Urban population (%)		0.022 (0.017)	0.033 (0.049)	0.013 (0.018)	−0.030 (0.050)	−0.024 (0.048)
Democracy index		0.018 (0.013)	−0.009 (0.012)	0.016 (0.013)	−0.008 (0.013)	−0.007 (0.012)
Life expectancy (years)		0.190*** (0.059)	0.011 (0.047)	0.186*** (0.058)	−0.076 (0.065)	−0.055 (0.065)
Female labor force (%)		0.079*** (0.027)	0.181*** (0.067)	0.074*** (0.027)	0.129* (0.073)	0.105 (0.070)
Women × Log GDP per capita						0.019* (0.010)
Constant	7.229*** (0.157)	−12.172*** (2.639)				
Observations	2868	2868	2868	2868	2868	2868
R2	0.102	0.390	0.882	0.404	0.887	0.888
Within R²			0.045	0.401	0.027	0.035
Country FE	No	No	Yes	No	Yes	Yes
Year FE	No	No	No	Yes	Yes	Yes
SE type	Het.-rob.	by state	by state	by state	by state	by state

* p <0.1, ** p <0.05, *** p <0.01